STA301 Statistics and Probability

Question No : 1 of 52

Analysis of variance is a procedure that enables us to test the equality of several

Answer (Please select your correct option)

- Variances
- Means
- Proportions
- Groups

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STA301 Statistics and Probability

Question No : 2 of 52

When two coins are tossed simultaneously, P (one head) is:

Answer (Please select your correct option)

- $\frac{1}{4}$
- $\frac{1}{2}$
- $\frac{3}{4}$
- 1

Statistics and Probability

Question No : 4 of 52

For testing of hypothesis about population proportion , we use:

Answer (Please select your correct option)

- Z-test
- t-Test
- Both Z & T-test
- F test

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4

Question Summary : (Alternate

STA301 Statistics and Probability

Question No : 3 of 52

An expected value of a random variable is equal to:

Answer (Please select your correct option)

- Variance
- Mean
- Standard deviation
- Covariance

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STA301 Statistics and Probability

Question No : 4 of 52

The curve of the F- distribution depends upon:

Answer (Please select your correct option)

- Degrees of freedom
- Standard deviation
- Mean
- Variance

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STA301 Statistics and Probability

Question No : 5 of 52

$E(4X + 5) =$ _____

Answer (Please select your correct option)

- 12 E (X)
- 4 E (X) + 5 ✓
- 16 E (X) + 5
- 16 E (X)

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5

Question No : 5 of 52

The degrees of freedom for a t-test with sample size 6 is:

Answer (Please select your correct option)

- 1
- 3
- 5
- 7

Question No : 6 of 52

The sum of deviations is zero, when deviations are taken from:

Answer (Please select your correct option)

- Mean
- Median
- Mode
- H.M

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6

Question No : 6 of 52

In the case of the sampling distribution of $\bar{X}$, the finite population correction factor (fpc) is:

Answer (Please select your correct option)

- $\sqrt{\frac{n-N}{N-1}}$
- $\sqrt{\frac{N-1}{N-n}}$
- $\sqrt{\frac{N-n}{N-1}}$
- $\sqrt{\frac{N-n}{N}}$

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6

Question Summary : (Attempted Question)

2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31

Question No : 7 of 52

Mark

If the probability of one event does not affected by the occurrence of another event then both events a

Answer (Please select your correct option)

- Dependent
- Not Mutually Exclusive
- Mutually Exclusive
- Independent



Question No : 6 of 52 Mar

In the case of the sampling distribution of $\bar{X}$, the finite population correction factor (fpc) is:

Answer (Please select your correct option)

- $\sqrt{\frac{n-N}{N-1}}$
- $\sqrt{\frac{N-1}{N-n}}$
- $\sqrt{\frac{N-n}{N-1}}$
- $\sqrt{\frac{N-n}{N}}$

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6

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33

STA301 Statistics and Probability

Question No : 7 of 52

When the random variable X and Y are independent then their co-variance is:

Answer (Please select your correct option)

- One
- Negative
- Zero
- Positive

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7

Question Summary : (Attempted Question)

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Question No : 8 of 52 Marks: 1 (Budgeted Tim

An estimator which has the smallest standard error among all unbiased estimators fulfills the property of _____.

Answer (Please select your correct option)

- Unbiasedness
- Efficiency
- Consistency
- Sufficiency

Question No : 9 of 52

If $\bar{X} = 5$ then which of the following expression is minimum?

Answer (Please select your correct option)

- $\Sigma(x-25)^2$
- $\Sigma(x-5)$
- $\Sigma|x-5|$
- $\Sigma|x-25|$

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STA301 Statistics and Probability

Question No : 10 of 52

For a positively skewed distribution m_3 will be:

Answer (Please select your correct option)

- Positive
- Negative
- Zero
- 1

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STA301 Statistics and Probability

Question No : 11 of 52

Pearson's coefficient of skewness is not possible if standard deviation is:

Answer (Please select your correct option)

- 1
- 1.5
- 0
- 3

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11

Question No : 9 of 52

Which one of the following assumptions is not a requirement for ANOVA?

Answer (Please select your correct option)

- Dependent samples
- Normal populations
- random samples
- Independent samples

Question No : 12 of 52

If a significance level of 1% is used rather than 5%, the null hypothesis is:

Answer (Please select your correct option)

- More likely to be rejected
- Less likely to be rejected
- Just as likely to be rejected
- None of the above

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Question No : 13 of 52

A judge can acquit a guilty person is the example of:

Answer (Please select your correct option)

- Type I error
- Type II error
- Correct decision
- No information regarding this

Question No : 14 of 52

Ideally, the width of confidence interval should be:

Answer (Please select your correct option)

- 0
- 1
- 99
- 100

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14

The degrees of freedom for a t-test with sample size 'n' is:

Answer (Please select your correct option)

- n-1
- n+1
- n-2
- n+2

Question No : 15 of 52

Marks: 1 (Budgeted Time Min)

If $\bar{X}$ is the mean of the n observations, then which test statistic will be used to calculate the confidence limits of the population variance σ^2?

Answer (Please select your correct option)

- Z-statistic
- T-statistic
- χ^2-statistics
- F-statistics

Question No : 16 of 52

If mean of χ^2 distribution is k then variance will be:

Answer (Please select your correct option)

- k^2
- 2k
- 1/k
- k

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16

Question No : 17 of 52

Mean of the F-distribution is possible only, when

Answer (Please select your correct option)

- $v_1 > 2$
- $v_2 > 2$
- $v_1 < 2$
- $v_2 < 2$

For the given poisson distribution $P(X=1)=\frac{e^{-0.135}0.135^1}{1!}$ the mean value is :

Answer (Please select your correct option)

- $e^{-0.135}$
- -0.135
- 0.135
- 1

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11

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34

Question No : 18 of 52

Marks:

As a rule, if p-value is _____ the level of significance, then we should reject Ho.

Answer (Please select your correct option)

- Equal to
- Less than
- Greater than
- Equal or greater than

STA301 Statistics and Probability

Question No : 19 of 52

Marks: 1 (Budgeted Time Mi

When the distance between observe frequency (o_i) and expected frequency (e_i) become larger, then the chi-square statistic become

Answer (Please select your correct option)

- Narrower
- Larger
- Zero
- Infinity

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An estimator which has the smallest standard error among all unbiased estimators fulfills the property of _____

Answer (Please select your correct option)

- Unbiasedness
- Efficiency
- Consistency
- Sufficiency

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12

Question No : 13 of 52

Marks: 1 (Budgeted Tim

When f (x, y) is bivariate probability density function of continuous random variables X and Y, then $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)dx\,dy$ to:

Answer (Please select your correct option)

- -1
- 0
- 1
- ∞

Question No : 20 of 52

In the test of goodness of fit, the _____ is used as a test statistic.

Answer (Please select your correct option)

- F
- t
- Z
- χ^2

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Which one is the formula for calculating the variance of the t-distribution?

Answer (Please select your correct option)

- $\sigma^2 = \sqrt{\frac{v}{v-2}}$
- $\sigma^2 = \frac{v^2}{v-2}$
- $\sigma^2 = \frac{v}{v-1}$
- $\sigma^2 = \frac{v}{v-2}$

STA301 Statistics and Probability

Question No : 21 of 52 **Marks: 1 (Budgeted Time Mi**

If there are 8 treatments with 6 blocks in a randomized completed block design then what are the degrees of freedom for treatments?

Answer (Please select your correct option)

- 5
- 4
- 6
- 7

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21

Question Summary : (Attempted Question)

Question No : 22 of 52 Marks: 1 (Budge

As the degree of freedom increases, the t-distribution tends to coincide with:

Answer (Please select your correct option)

- Binomial distribution
- Uniform distribution
- Hypergeometric distribution
- Normal distribution

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22

Question Summary : (Attempted Question)

Which one is the correct formula to find the desired sample size?

Answer (Please select your correct option)

- $n = \left(\frac{Z_{\alpha/2} \cdot \sigma}{e} \right)^2$ ✓
- $n = \left(\frac{Z_{\alpha/2} \cdot \sqrt{\sigma}}{e} \right)^2$
- $n = \left(\frac{Z_{\alpha/2} \cdot \bar{X}}{e} \right)^2$
- $n = \frac{Z_{\alpha/2} \cdot \sigma}{e}$

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16

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 3

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STA301 Statistics and Probability

Question No : 22 of 52

As the degree of freedom increases, the t-distribution tends to coincide with:

Answer (Please select your correct option)

- Binomial distribution
- Uniform distribution
- Hypergeometric distribution
- Normal distribution

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22

Question Summary : (Attempted Question)

To find the confidence interval for the ratio of two variances, we use

Answer (Please select your correct option)

- F-Distribution
- Z-Distribution
- Chi-square-Distribution
- t-Distribution

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17

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30

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Question No : 23 of 52

What factor determines the shape of the t-distribution?

Answer (Please select your correct option)

- Degree of freedom
- Critical value
- Frequency of data
- Probability

In case of a 3×3 contingency table what is the value of degrees of freedom?

Answer (Please select your correct option)

- 2
- 4
- 6
- 9

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18

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32

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Question No : 24 of 52 Marks: 1 (Budgeted Time

If all the values fall on the same straight line and the line has a positive slope then what will be the value of the correlation coefficient 'r'?

Answer (Please select your correct option)

- $0 \leq r \leq 1$
- $r \geq 0$
- $r = +1$
- $r = -1$

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24

Question No : 25 of 52

Marks: 1 (Budgeted Time Min)

In a one way ANOVA test there are 5 observations in each of three treatments. The degrees of freedom for the treatments is:

Answer (Please select your correct option)

- 5
- 3
- 1
- 2

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25

Question Summary : (Attempted Question)

Question No : 28 of 52

Marks: 1 (Budgeted Time Min)

In a set of 10 observations the mean is 20 and the median is 15. There are 2 values that are 6, and all other values are different. What is the value of mode:

Answer (Please select your correct option)

- 15
- 6
- 10
- 2
- 6

Question No : 21 of 52 Marks: 1

In statistics LSD is the abbreviation of.....

Answer (Please select your correct option)

- Large sample difference
- Least significant difference
- Least significant digit
- Low speed data

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21

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 3

Question No : 31 of 52

Let X and Y are two random variables. Which property of expectation is true?

Answer (Please select your correct option)

- $E(2X - 3Y) = 2E(X) - 3E(Y)$
- $E(2X - 3Y) = 2E(X) - 3E(Y) + E(XY)$
- $E(2X - 3Y) = 2E(X) + 3E(Y) - E(XY)$
- $E(2X - 3Y) = 2E(X) + 3E(Y) - 2\,E(XY)$

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31

Question No : 22 of 52 Marks

As the degree of freedom increases, the t-distribution tends to coincide with:

Answer (Please select your correct option)

- Binomial distribution
- Uniform distribution
- Hypergeometric distribution
- Normal distribution ✓

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22

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34

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If we plot the paired observation (X_i, Y_i) $i = 1, 2, \ldots, n$ on a graph, the resulting set of points is called:

Answer (Please select your correct option)

- Frequency polygon
- Frequency histogram
- Scatter diagram
- Cumulative frequency polygon

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23

Question Summary : (Attempted Question)

STA301 Statistics and Probability

What is the result of the following factorial expression?

$1! + 2! + 3!$

Answer (Please select your correct option)

- 5
- 6
- 9
- 10

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24

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36

Question No : 25 of 52

When f(x) is continuous probability function, then P(X = 2) is:

Answer (Please select your correct option)

- 1
- 0.5
- 0 ✓
- 0.25 ✗

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25

Question Summary : (Attempted Question)

Question No : 26 of 52

Marks:

All possible trials by throwing a die, is an example of:

Answer (Please select your correct option)

- Finite population
- Infinite population
- Hypothetical population
- Limited population

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26

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36

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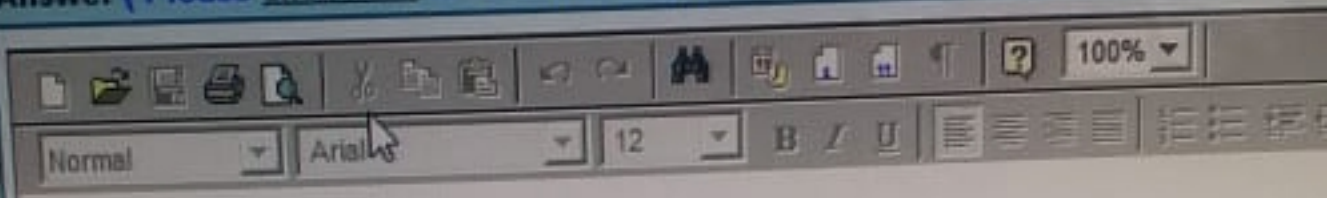
Question No : 52 of 52

The continuous random variable X having the p.d.f

$f(x) = 2(1-x), \qquad 0 < x < 1$

Using the integration, find the expected value $E(X^4)$

Answer (Please click here to Add Answer)



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52

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29

STA301 Statistics and Probability

Question No : 33 of 52

Parameters of the hyper geometric distribution h(x;N,n,k) are:

Answer (Please select your correct option)

- X, N, k
- N, n, k
- X, n, k
- X, N, n, k

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33

Question Summary : (Attempted Question)

Question No : 35 of 52

Marks: 1 (Budgeted Time Min)

If the population standard deviation σ is unknown and the sample size n is less than or equal to 30, the confidence interval for the population mean μ is:

Answer (Please select your correct option)

- $\hat{p} \pm Z_{\alpha/2} \sqrt{\frac{\hat{p}\hat{q}}{n}}$
- $\bar{X} \pm Z_{\alpha/2} \frac{s}{\sqrt{n}}$
- $\bar{X} \pm t_{\alpha/2(\nu)} \frac{s}{\sqrt{n}}$
- $\bar{X} \pm t_{\alpha/2(\nu)} \frac{S_d}{\sqrt{n}}$

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35

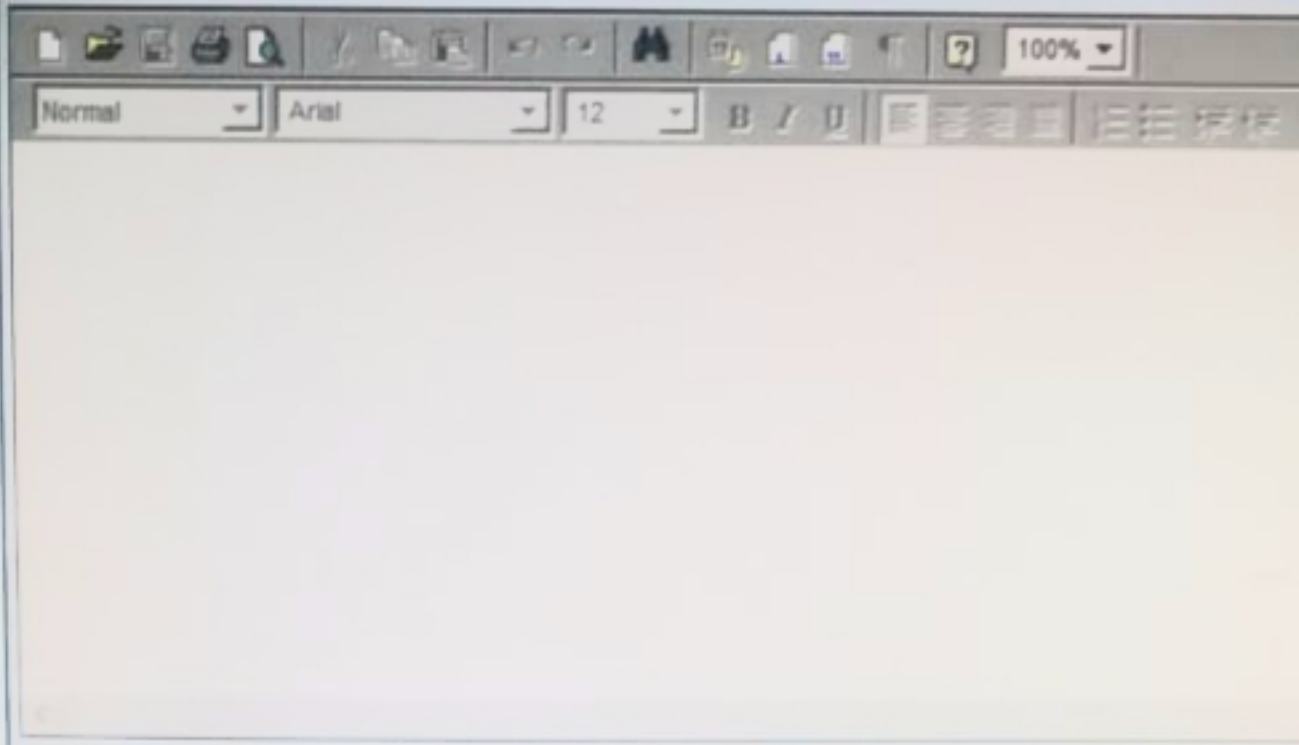
Question Summary : (Attempted Question)

Question No : 51 of 52

Find the cumulative frequencies.

Age X	No. of Students f	C.F
13	6	
..	..	

Answer (Please click here to Add Answer)



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51

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30

Question No : 36 of 52

Marks: 1 (Budgeted Time

For a normal distribution with $\mu=55$ and $\sigma=10$, how much area will be found under the curve to the right of $X=55$?

Answer (Please select your correct option)

- 1.0
- 0.68
- 0.5
- 0.32

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36

Question No : 38 of 52

Marks: 1 (Budgeted Time Min)

Suppose that, for a certain population, $\sigma_{\bar{x}}$ is calculated as 20, when samples of size 25 are taken. What must be the values of σ for this infinite population?

Answer (Please select your correct option)

- 1000
- 500
- 100
- 4

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38

Question Summary : (Attempted Question)

Question No : 41 of 52

Marks: 2 (Budgeted Time Min)

Suppose we want to show that only children have an average higher cholesterol level than the national average. It is known that the mean cholesterol level for all Americans is 190. Construct the relevant hypothesis test.

Answer (Please click here to Add Answer)

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41

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43

STA301 Statistics and Probability

Question No : 42 of 52

Calculating the value of χ^2 test statistics using the given information:

n=25, S^2=3, and σ_0^2=18

Answer (Please click here to Add Answer)

Normal | Arial | 12 | 100%

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42

Question Summary : (Attempted Question)

STA301 Statistics and Probability

Question No : 44 of 52

Marks

Solve the *Combination (With formula)*

${}^{5}C_{2} =$

Answer (Please click here to Add Answer)

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Question No : 45 of 52

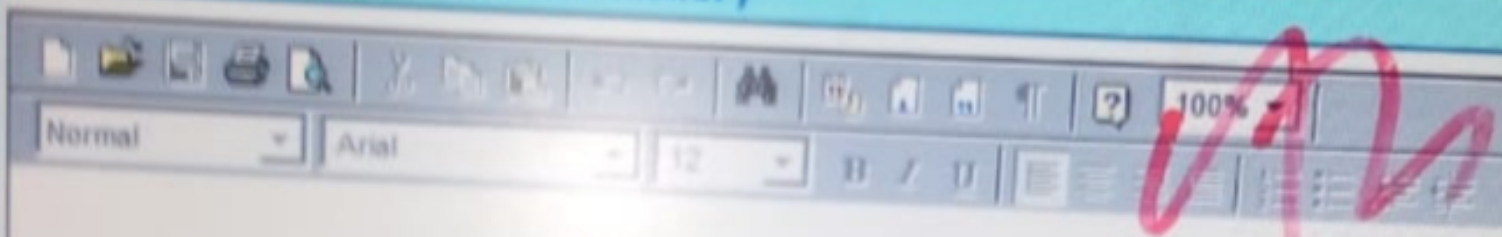
Construct 90% confidence interval for σ^2 from the following information,

$\sum(X-\bar{X})^2 = 25,$

$\chi^2_{0.05(7)} = 14.07$

$\chi^2_{0.95(7)} = 2.17$

Answer (Please click here to Add Answer)



Let suppose $\hat{P}_1 = 0.4$, $n_1 = 10$ *and* $\hat{P}_2 = 0.3$, $n_2 = 20$ *then* using 0.05 significance level and critical value as $Z < -Z_{0.05} = -1.645$, test the following

$H_0 : P_1 - P_2 \geq 0.01$

$H_1 : P_1 - P_2 < 0.01$

Answer (Please click here to Add Answer)

Marks: 5 (Budgete

The accompanying data describe the weights (in grams) of 24 electronic devices.

9.2	9.7	8.8	10.7	8.4	8.7	10.7	6.9	8.2	8.3	7.3	9.1
7.8	8.0	8.6	7.8	7.5	8.0	7.3	8.9	10.0	8.8	8.7	12.6

Calculate a *point estimate* of the mean value of weights of the electronic device. State which estimator you used.

Answer (Please click here to Add Answer)

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49

Question Summary : (Attempted Question)

Average mark of the students in a class is 35. A sample of 9 students showed the following marks.

45 32 37 46 39 36 41 48 36

Calculate sampling error?

Answer (Please click here to Add Answer)

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Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32

STA301 Statistics and Probability

Question No : 47 of 52

Marks: 3 (Budgeted Time Min)

A fair coin is tossed 25 times. Applying normal approximation to the binomial distribution, find the mean and variance.

Answer (Please click here to Add Answer)

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47

Question Summary : (Attempted Question)

A 99% confidence interval fot the population mean μ is determined to be(65.32 to 73.54). If the confidence level is re 90% the confidence interval forμ

Answer (Please select your correct option)

- Become wider
- Become narrower
- Remains unchanged
- None of these

All hypothesis tests start with

Answer (Please select your correct option)

- A null hypothesis
- Alternative hypotheses
- Test statistic
- Type-I Error

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32

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34

The variance of a binomial distribution depends on:

Marks: 1 (Budg

Answer (Please select your correct option)

- Number of trials
- Probability of success
- Probability of failure
- All of the above

The total number of samples when sampling is done without replacement is equal to:

Answer (Please select your correct option)

- N^n
- C_n^N
- $\frac{N-n}{N-1}$
- 1

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Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40

STA301 Statistics and Probability

Question No : 39 of 52

Which one is commonly called a bell shaped distribution?

Answer (Please select your correct option)

- Symmetrical
- Bimodal
- Skewed
- U shaped

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Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34

Question No : 40 of 52

We take ..... on y axis to construct a frequency polygon.

Answer (Please select your correct option)

- (•) Class frequencies
- ( ) Mid points
- ( ) Class limits
- ( ) Class boundaries